

Demonstrating FIRE: Mid-Air Thermo-Tactile Display

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Figure 1: (a) User interacting with the system while experiencing (b) campfire, (c) water fountain, (d) steam, and (e) candlelight.

ABSTRACT

We present an innovative mid-air thermo-tactile display system based on ultrasound haptics. The system features an open-top chamber, heat modules, and an ultrasound haptic display. Our approach directs heated airflow toward the pressure point created by the ultrasound display, delivering thermal and tactile sensations simultaneously in mid-air. We demonstrate our system in four distinct VR environments—*campfire*, *water fountain*, *kitchen*, and *candle*—highlighting the enhanced user experiences made possible by the integration of thermal and tactile feedback.

KEYWORDS

Thermal Feedback, Mid-Air Haptics, Virtual Reality

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1 INTRODUCTION

In recent years, significant emphasis has been placed on incorporating thermal feedback in VR applications [5, 6] to enhance immersion and presence. Thermal feedback serves as an ambient communication channel, contributing to a more realistic perception of virtual environments. Just as in the physical world, many VR scenarios benefit from thermal feedback—users can feel the warmth of a fire in a virtual fireplace, experience the sensation of hot water from faucets or showers, or perceive the chill of falling snow.

Numerous studies on non-contact thermal and humidity feedback have employed various methods, utilizing thermal devices such as heaters, blowers, projector lights, and infrared lamps. For instance, Hülsmann et al. [1] provided wind and thermal feedback using IR lamps and fans within the interaction space. Motoyama et al. [3] used mist to create cold sensations on the fingertip, while Kamigaki et al. [2] proposed an ultrasound display to generate a focal point in mid-air on a hand-worn glove to deliver thermal sensations. However, these approaches often struggle to control thermal cues in 3D space due to their underlying mechanisms. Additionally, while these methods offer better control, they typically do not support bare-hand interaction.

Our perception-based approach integrates localized ultrasound tactile feedback with global thermal feedback, creating a thermo-tactile sensation on bare hands in mid-air. Humans perceive thermal and tactile information through different mechanisms: tactile stimuli are detected by mechanoreceptors and processed in the

somatosensory cortex, while thermal stimuli are detected by thermoreceptors and processed in the insula cortex. When these two types of stimuli are applied to the same area of the hands without masking each other, they are perceived as a unified thermo-tactile feedback. This allows for the presentation of thermo-tactile cues in 3D space at arbitrary locations. It combines vibrotactile and thermal stimuli without interference by generating ambient thermal cues through a heated chamber. By presenting acoustic pressure points alongside these ambient thermal cues, we achieve integrated thermo-tactile sensations in mid-air for bare-hand interaction. Visual information is closely coupled with these thermo-tactile cues (see Figure 1). Further details about this system and the experiments conducted to validate our approach can be found in [4].

2 SYSTEM

2.1 Hardware

In Figure 2, our setup delivers thermo-tactile feedback in mid-air, using an ultrasound haptic display (STRATOS Explore, Ultraleap) to provide tactile stimuli at 200 Hz to the user's palm, while the Meta hand-tracking system tracks hand movements. Thermal feedback is managed by a PTC heating element, which adjusts its temperature through resistance changes, controlled by a digital power supply and a microcontroller. The hexagon-shaped chamber, built with aluminum beams and insulation foam, ensures effective thermal flow. This system integrates with an Oculus Quest 2 VR headset.

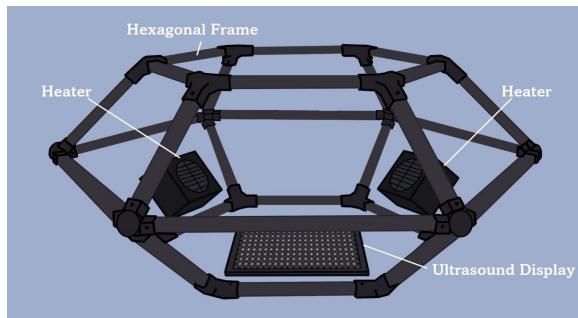


Figure 2: Labeled diagram of the proposed system

2.2 Visual Scenes

We demonstrate our approach using four VR scenes, each providing unique thermo-tactile feedback. We describe each environment in detail below:

Campfire: Figure 1 (b) shows the campfire scene. In this environment, the user interacts with fire embers and rising smoke with their virtual hands. An ultrasonic cue at 200 Hz with 0.7 intensity for haptic rendering is provided for 500 ms whenever an ember collides with the virtual hands. The temperature for this scene is set to 44°C, which maximizes the heat effect without reaching the pain threshold, simulating the warmth of a fire. A crackling fire sound is played to enhance the user experience.

Water Fountain: In this scene, a hot water fountain with a water stream is presented in a park (see Figure 1 (c)). When the water

stream collides with the user's hands, the height of the water stream dynamically adjusts depending on the hand's position. A continuous circular haptic sensation with a 2 cm radius, frequency of 200 Hz, and intensity of 1.0 is presented to the user's palm. A temperature of 38°C is selected to emulate the sensation of warm water hitting the palm.

Kitchen: In this scene, users find themselves in a kitchen with a teapot on the stove, emitting steam (see Figure 1 (d)). Participants can interact with the steam using their virtual hands. A haptic cue with a 4 cm radius, 200 Hz frequency, and 0.4 intensity is presented on the user's palm to simulate the sensation of steam interacting with the skin. The temperature for this scene is set to 40°C to replicate the warmth of steam.

Candle: In this scene, a candle with a bright flame is placed on a desk (see Figure 1 (e)). Participants can place their hands above the candle flame to perceive the fluctuating hot air. A circular haptic cue sensation with a radius changing from 2 cm to 0.5 cm over 2 seconds, a frequency of 200 Hz, and an intensity change from 1.0 to 0.8 is provided at 25 cm above the ultrasonic board. A temperature of 36°C is chosen to simulate the flickering of candle flame.

3 DEMO EXPERIENCE

At UIST 2024, we will showcase the proposed thermo-tactile display within a VR setup. Attendees interested in the interactive demo will receive a briefing on how to interact with the system. They will then stand in front of the system and put on the headset. Demonstrators will perform a calibration process to synchronize the attendee's physical and virtual world positions. Following calibration, attendees can begin interacting with the prototype, engaging with all the interactive objects. The entire demo, including calibration and briefing, will take no more than 5 minutes per attendee.

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